# Topic 7 — Wildlife Conservation (वन्यजीव संरक्षण)

**Covers syllabus (click to expand)**

Wildlife Conservation | Wildlife Protection (वन्यजीव सुरक्षा) | Project Tiger (प्रोजेक्ट टाइगर) | Project Elephant (प्रोजेक्ट एलीफेंट) | Crocodile Conservation Project | Asiatic Lion (सिंह) Conservation | Rhino Conservation | Snow Leopard Conservation | Vulture Conservation

**Sources baked in:** WPA 1972 (amended 2006), MoEFCC, NTCA, BNHS, IUCN, NCERT Biology Ch 15–16, UPPCS PYQs 2018–2025 **Weight:** ★★★ High — launch years, statutory vs executive traps, species ↔ habitat **Last verified:** August 2026

## Current Affairs (this topic)

- Track Project Tiger / Project Elephant / species project headlines against the statutory vs scheme trap.
- NTCA tiger-reserve count drifts — teach the concept; verify the live total before a count stem.

## 7.1 Wildlife Conservation

**Identity:** In-situ primary | Species programmes + community participation | MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) nodal

- **Wildlife conservation** means protecting wild plants and animals and the habitats they need to survive.
- The main strategy is **in-situ conservation** (स्व-स्थाने संरक्षण), which protects species where they naturally live.
- **Ex-situ conservation** (बाह्य-स्थाने संरक्षण) uses zoos, captive breeding, and gene banks as backup when wild populations are critically low.
- A **flagship species** (प्रतीक (फ्लैगशिप) प्रजाति) such as the tiger attracts funding and public attention for conservation.
- An **umbrella species** (छत्र (अंब्रेला) प्रजाति) such as the tiger or elephant needs large habitat, so protecting it also protects many other species.
- A **keystone species** (कीस्टोन प्रजाति) such as the vulture plays an irreplaceable ecological role. Its loss harms the whole system.
- Major threats include habitat fragmentation, poaching, human–wildlife conflict, veterinary drug poisoning, and invasive species.
- India runs **species-specific programmes** under MoEFCC on top of the protected area network.
- **World Wildlife Day** is observed on **3 March**, marking the adoption of CITES (साइट्स) in 1973.
- **Wildlife Week** in India is observed from **2 to 8 October**.

- **Operation Olivia** is an **Indian Coast Guard** initiative to protect **Olive Ridley turtles** along the **Odisha coast**.
- Olive Ridley **nesting season** runs from **November to February**, when mass **arribada** occurs at Gahirmatha and Rushikulya.

**PYQ — UPPCS Prelims 2025, Q62**

Which pairs is/are NOT correctly matched? (Day) — (Date): 1. Ozone Day — 16 Sept 2. WED — 5 June 3. World Wildlife Day — 22 March

A. 1 and 2

B. Only 3

C. 2 and 3

D. Only 1

**Show answer**

**Ans: B** — World Wildlife Day = **3 March**.

**PYQ — UPPCS Prelims 2025, Q143**

With reference to Operation Olivia, which is/are correct?

1. ICG initiative to protect Olive Ridley turtles along Odisha coast.
2. Carried out in August during nesting.

A. Only 2

B. Neither

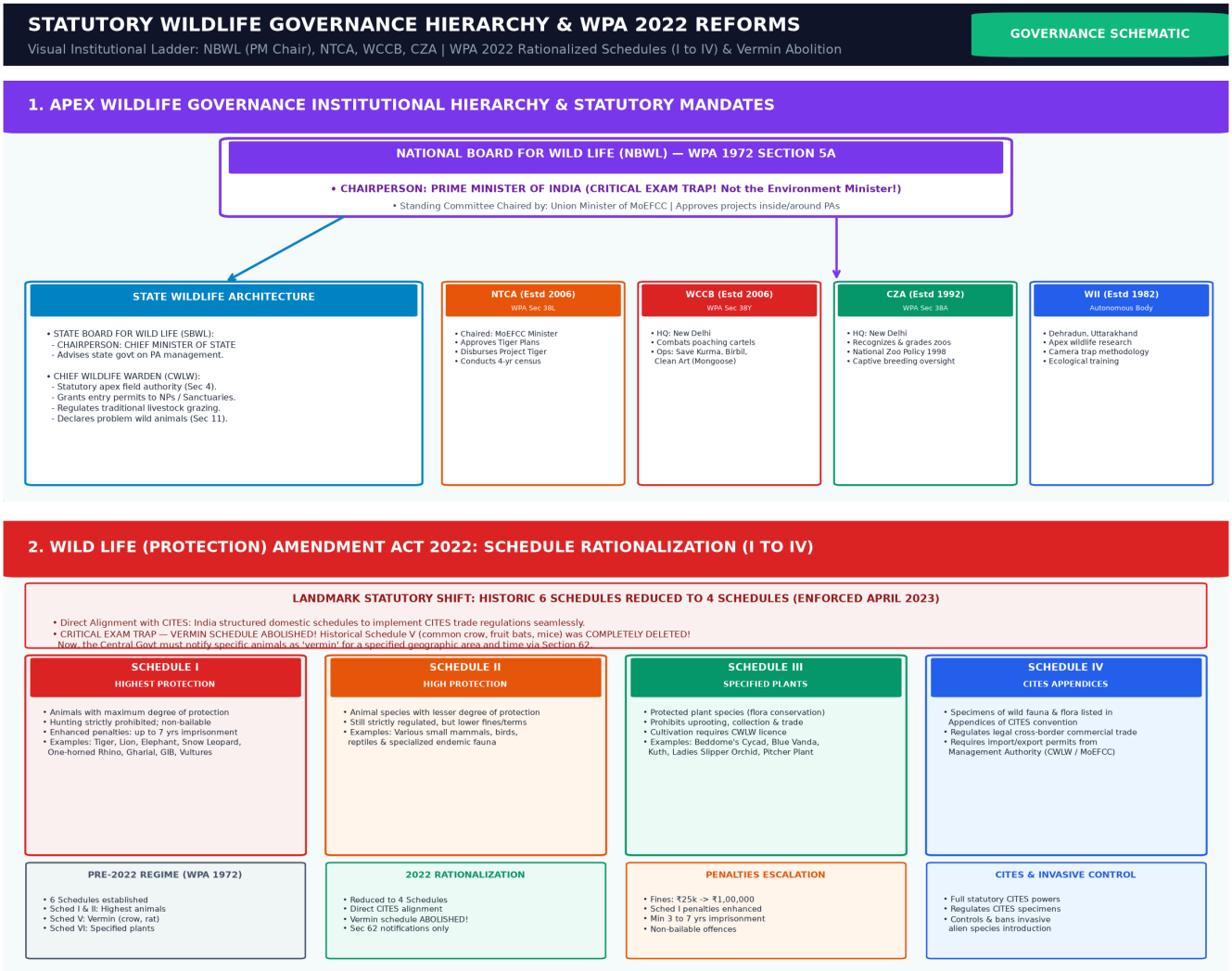
C. Both

D. Only 1

**Show answer**

**Ans: D** — Nesting is **Nov–Feb**, not August.

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## 7.2 Wildlife Protection

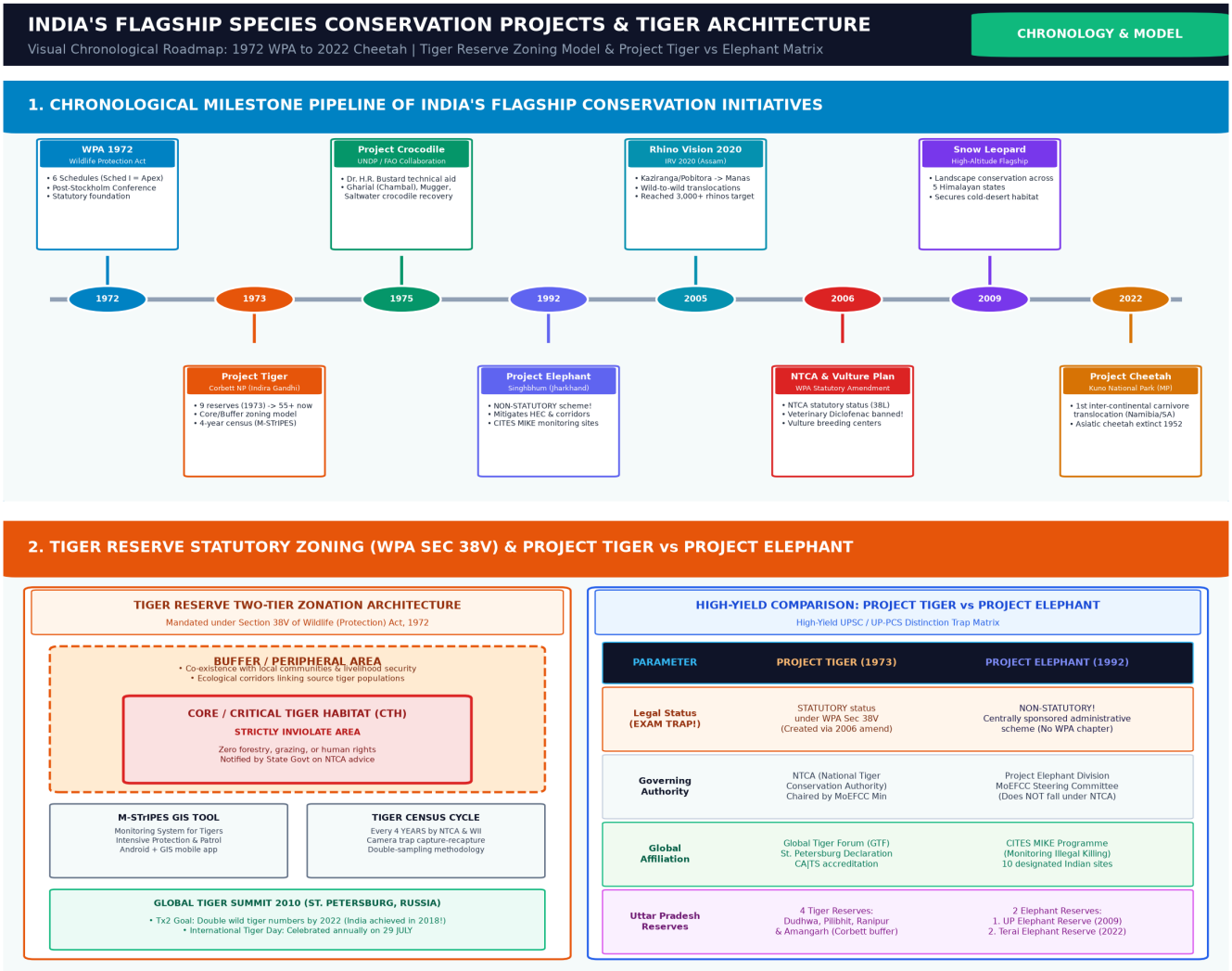
**Identity:** WPA 1972 | 6 Schedules | CWLW + NBWL | Art 48A, 51A(g)

- **Wildlife (Protection) Act, 1972** is India's main law for protecting wild animals, birds, and plants.
- The Act was enacted in the same year as the **Stockholm Conference (1972)** (स्टॉकहोम सम्मेलन).
- The Act divides species into **six schedules** based on protection level.
- **Schedule I** gives maximum protection. Hunting and trade (पण्याध्यक्ष) are banned for tiger, elephant, rhino, lion, gharial, snow leopard, and Indian vultures.
- If a tortoise (or any species) is placed in **Schedule I**, it enjoys the **same protection level as the tiger**.
- **Schedule V** lists vermin such as the common crow and rat, which may be hunted.
- **Schedule VI** lists specified plants whose cultivation is **prohibited except with a licence** from the Chief Wildlife Warden or authorised officer (classic pitcher-plant / specified-plant pattern).
- Gharial, Indian wild ass, and wild buffalo cannot be hunted except under limited WPA provisions.
- The **Chief Wildlife Warden (CWLW)** is the state-level authority for permits and sanctuary management.
- The **National Board for Wildlife (NBWL)** (राष्ट्रीय वन्यजीव बोर्ड) is chaired by the **Prime Minister** and advises on protected area policy.

- Offences against Schedule I species can lead to imprisonment up to **7 years**.
- **Article 48A** directs the State to protect wildlife. **Article 51A(g)** makes it a citizen's duty to protect wildlife.
- Once an area is notified as a **Community Reserve**, hunting is banned, NTFP collection may continue, but **traditional agriculture / jhum is not** allowed; CWLW authority under WPA applies.
- **Animal Welfare Board of India** was established in **1962** under the **Prevention of Cruelty to Animals Act, 1960** — not under EPA 1986.

| Schedule | Protection | Examples |
|---|---|---|
| **I** | Maximum — hunting banned | Tiger, elephant, rhino, gharial, vulture |
| **II** | High | Himalayan black bear, cobra |
| **III–IV** | Moderate (नरम दल)/lower fauna (प्राणीजात) | Chital, sambar, hare |
| **V** | Vermin — may be hunted | Crow, rat |
| **VI** | Specified plants — licence needed | Pitcher plant and other listed taxa |

> **Logic:** WPA = **1972**. Project Tiger = **1973**. Do not swap the two years. Schedule VI ≠ absolute cultivation ban.




## 7.3 Project Tiger

**Identity:** CSS 1973 | NTCA 2005/2006 | 58 TRs | Umbrella species model

- **Project Tiger** was launched in **April 1973** by PM Indira Gandhi (गांधी) at Jim Corbett (जिम कॉर्बेट) National Park.
- The project began with **9 tiger reserves**. **Corbett** was the first reserve.
- India now has **58 tiger reserves**. **Madhav Tiger Reserve** (MP) is the 58th.
- The **National Tiger Conservation Authority (NTCA)** became a statutory body under the **WPA 2006 amendment**.
- Every tiger reserve must have a **core zone** (कोर क्षेत्र) kept inviolate and a **buffer zone** (बफर क्षेत्र) for regulated use.
- The **All India Tiger Estimation** runs every **4 years** using camera traps and the **M-STrIPES** app (Monitoring System for Tigers — Intensive Protection and Ecological Status: field patrol protocols + GIS software for management).
- **International Tiger Day** is **29 July** (decided at the **2010 St Petersburg** Tiger Summit).
- **Nagarjunsagar–Srisailam** holds the largest critical tiger habitat among common options. **Bandipur** (बांदीपुर) pioneered drone monitoring among common stems.

- **Sathyamangalam TR (Tamil Nadu (तमिलनाडु))** sits at the Eastern–Western Ghats (पश्चिमी घाट) junction.
- The **2022 census** recorded **3,682 tigers** in India.
- **Madhya Pradesh** has the most tiger reserves among states.
- India hosts about **70% of the world's wild tigers**.
- The tiger is an **umbrella species**. Protecting it also protects prey, co-predators, and forest habitat.
- UP tiger reserves are **Dudhwa (1987)**, **Pilibhit (2014)**, and **Ranipur (2023)**.

> **Logic:** Tiger Reserve has **WPA statutory basis (Sec 38V)**. Elephant Reserve (हाथी अभयारण्य) does **not**.

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## 7.4 Project Elephant

**Identity:** CSS 1992 | 33 ERs | 150 corridors | No WPA statutory basis

- **Project Elephant** was launched in **1992** as a Centrally Sponsored Scheme.
- It aims to protect elephants, restore corridors, and reduce **human–elephant conflict**.
- It covers welfare of **wild Asian elephants and captive elephants** — “wild only” is a false stem.
- **Elephant Reserves** are notified by state governments on MoEFCC recommendation.
- India has **33 Elephant Reserves** across **14 states**.
- About **150 elephant corridors** have been identified for genetic exchange between herds.
- Elephant Reserves have **no separate statutory category under WPA 1972**.
- Legal protection inside an ER comes from overlapping sanctuaries, national parks, and forest laws.
- India's elephant population is estimated at about **30,000** (2020 census trend).
- UP has **Uttar Pradesh Elephant Reserve (2009)** and **Terai Elephant Reserve (2022)**.
- The Asian elephant is listed in **WPA Schedule I**.

> **Logic:** Trap — "Elephant Reserve is declared under WPA like Tiger Reserve" is **false**.

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## 7.5 Crocodile Conservation Project

**Identity:** 1975 | Crocodile Breeding and Management Project | 3 Indian species

- The **Crocodile Conservation Project** was launched in **1975** with UNDP (यूएनडीपी)/FAO assistance.
- Its official name is the **Crocodile Breeding and Management Project**.
- It covers **three species**: **gharial**, **mugger (marsh crocodile)**, and **saltwater crocodile**.
- The **gharial** has a long narrow snout and lives in clean deep rivers such as the **Chambal** and **Girwa**.
- To see gharials in a natural habitat, **Chambal River / National Chambal Sanctuary** is the standard answer among common options.
- The **mugger** is the most widespread Indian crocodilian and occurs in rivers, lakes, and tanks.

- The **saltwater crocodile** is the largest reptile and lives in coastal mangroves, notably at **Bhitar Kanika** (Odisha).
- The gharial is **Critically Endangered** on the IUCN Red List.
- **National Chambal Sanctuary** (UP–MP–Rajasthan) is India's main gharial stronghold.
- **Katarniaghat Wildlife Sanctuary** in Bahraich, UP, protects gharial on the Girwa river.
- **Whale shark** is India's (and among the world's) largest fish; seasonal aggregations off the **Gujarat** coast (about March–May); a dedicated protection programme began around **2004**.
- **Kharai camel** of Kutch can swim several kilometres in seawater and graze mangroves; it can be domesticated (statement "lives only wild / cannot be domesticated" is false).
- **Pugmark** technique estimates wild animal numbers from species-specific footprints.
- **Spider monkeys** occur in Central and South American tropical forests — they are **not** naturally found in India.
- Scientific name locks: Asiatic wild ass (**Equus hemionus**), Barasingha (**Rucervus/Cervus duvaucelii**), Chinkara (**Gazella bennettii**), Nilgai (**Boselaphus tragocamelus**).
- WWF has used the **Giant Panda (*Ailuropoda melanoleuca*)** as its logo since **1961**.
- The first International Tiger Forum / World Tiger Summit (**2010**) was held at **Saint Petersburg**, Russia.
- Captive breeding centres operate at Nandankanan, Hyderabad (हैदराबाद), Chennai, Lucknow (लखनऊ), and Trivandrum.
- **Ganges River Dolphin** (*Platanista gangetica*) is India's **National Aquatic Animal (notified 2009)** — one of four (चातुर्याम) obligate freshwater dolphins; Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र)–Meghna systems. Decline drivers: dams/barrages, fishing nets, agrochemicals — not mainly rising crocodile numbers. UP also uses a state aquatic-animal / Meri Ganga Meri Dolphin awareness framing.
- IUCN teaching: gharial often **Critically Endangered**; mugger/saltwater crocodile commonly **Vulnerable** in bank notes.

> **Logic:** The crocodile project covers **three species**, not gharial alone. National aquatic animal = **dolphin**, not gharial.

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## 7.6 Asiatic Lion Conservation

**Identity:** Gir, Gujarat only | Project Lion 2020 | Kuno second home planned

- The **Asiatic lion** (*Panthera leo persica*) survives in the wild only in the **Gir landscape of Gujarat**.
- India's wild Asiatic lion population is about **750+** (2025 trend).
- Historically lions ranged from Greece to Bihar. Hunting reduced them to Gir by the early 20th century.
- **Gir Conservation Programme** focused on habitat, prey base, and disease control after a **canine distemper** outbreak in 2018.
- **Project Lion** was launched in **August 2020** to strengthen Gir management and create a second population.

- **Kuno-Palpur Wildlife Sanctuary** in Madhya Pradesh is the planned second home for translocated lions.
- The Asiatic lion is in **WPA Schedule I** and is the **state animal of Gujarat**.
- **Uttar Pradesh has no wild Asiatic lion**. Dudhwa holds tiger and rhino, not lion.

> **Logic:** Trap — "Asiatic lions in Dudhwa/UP" is **false**. Gir = Gujarat only.

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## 7.7 Rhino Conservation

**Identity:** Greater one-horned rhino | Kaziranga (काजीरंगा) largest | IRV 2020 | Dudhwa UP site

- The **Greater One-Horned Rhinoceros** (*Rhinoceros unicornis*) lives in Terai and Brahmaputra floodplain grasslands.
- **Kaziranga National Park** (Assam) holds the **largest rhino population** (~2,600+).
- Poaching for horn remains the main threat despite strict anti-poaching in Kaziranga.
- **Indian Rhino Vision 2020 (IRV 2020)** aimed to spread rhinos across **7 protected areas**.
- Target sites include Kaziranga, Pobitora, Orang, and Manas in Assam, Jaldapara and Gorumara in West Bengal, and **Dudhwa** in UP.
- **Dudhwa National Park** in Lakhimpur Kheri received **reintroduced rhinos** under IRV 2020.
- India hosts over **80% of the global** greater one-horned rhino population.
- The rhino is **WPA Schedule I** and **IUCN Vulnerable**.

> **Logic:** UP rhinos are in **Dudhwa**, not Kaziranga. Kaziranga is in Assam.

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## 7.8 Snow Leopard Conservation

**Identity:** Project Snow Leopard 2009 | GSLEP 2013 | Himalaya only | 500–700 in India

- The **snow leopard** (*Panthera uncia*) lives at **3,000–5,400 m** in the Himalayas and trans-Himalayas.
- India has an estimated **500–700 snow leopards** across Ladakh (लद्दाख), HP, Uttarakhand (उत्तराखंड), Sikkim, Arunachal Pradesh, and J&K.
- **Project Snow Leopard** was launched in **2009** to strengthen high-altitude protected area management.
- **GSLEP (Global Snow Leopard & Ecosystem (पारिस्थितिकी तंत्र) Protection Programme)** was launched in **2013** by **12 range countries**.
- **Hemis National Park** (हेमिस) in Ladakh is a key snow leopard habitat.
- **SECURE Himalaya** is a UNDP-GEF project for high-altitude biodiversity in Himachal (हिमाचल), Uttarakhand, Sikkim, and Ladakh.
- The snow leopard is an **umbrella species** for alpine ecosystems.
- **International Snow Leopard Day** is **23 October**.

- The snow leopard is **WPA Schedule I**.

> **Logic:** Snow leopard is **not** found in Gir, Western Ghats, or Terai plains.

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## 7.9 Vulture Conservation

**Identity:** Diclofenac crisis | Vet ban 2006 | Action Plan 2006 | Pinjore breeding

- **Vultures** are keystone scavengers that quickly remove carcasses and limit disease spread.
- India lost about **99% of its vultures** between the 1980s and 2007 — one of the fastest avian declines on record.
- **BNHS research (2003)** identified **diclofenac**, a veterinary NSAID given to cattle, as the main cause of vulture renal failure.
- **Veterinary diclofenac was banned in 2006**. **Meloxicam** (मेलोक्सिकैम) is promoted as a safe alternative.
- The **Action Plan for Vulture Conservation 2006** has four pillars: ban toxic drugs, breeding centres, Vulture Safe Zones, and monitoring.
- **Vulture Safe Zones** are roughly **100 km radius** areas where diclofenac use is eliminated.
- Captive breeding centres operate at **Pinjore** (Haryana), **Rajabhatkhawa** (West Bengal), **Rani** (Assam), and **Bhopal** (MP).
- The **SAVE programme** (Saving Asia's Vultures from Extinction) coordinates international vulture recovery.
- Key Indian vulture species — white-rumped, long-billed, and slender-billed — are **Critically Endangered**.

> **Logic:** UPPCS 2018 pattern — cattle painkiller linked to vulture decline is **diclofenac**.

**PYQ — UPPCS Prelims 2018, Q131**

Which painkiller given to cattle is responsible for near extinction of vultures in India?

A. Ibuprofen

B. Acetaminophen

C. Aspirin

D. Diclofenac

**Show answer**

**Ans: D**

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